

Prescribed Dynamic Sheath Versus Static Vacuum Jacket in Three-Dimensional Plasma Fluid FDTD: Measurable Dynamic Feed Loading

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Key points

- A time-varying depleted radius is imposed in three-dimensional plasma fluid FDTD without particle-in-cell.
- Oscillating sheath cases sit far outside static thickness brackets at 0.5–1.2 MHz (soft updates).
- Feed $\arg(Z)$ tracks kinematic boundary phase over $\sim 70^\circ$; amplitude scaling with sheath swing remains unsettled.

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Abstract

Radio-frequency antennas in plasma are surrounded by a depleted near-zone layer whose thickness need not be frozen over a drive cycle. Most three-dimensional fluid electromagnetic models either omit that layer or treat it as a static vacuum jacket. Here we keep the Maxwell–fluid core fixed and prescribe a kinematic sheath radius $r_s(t) = r_{s0} + \Delta r \sin(\omega t + \phi)$, comparing feed diagnostics to static jackets at the mean and extreme thicknesses. At 700 kHz ($f_p = 2$ MHz), hard-staircase and soft-edge updates both place the oscillating phasor impedance far outside the static bracket segment in the complex plane. Varying the soft blend width over 0.5–2 cells moves the oscillating point but keeps it out-of-bracket. A soft sparse frequency set at 0.5, 0.7, and 1.2 MHz likewise remains out-of-bracket at all three tones. A phase sweep of the prescribed motion rotates feed $\arg(Z)$ by tens of degrees while staying out-of-bracket. A soft-edge Δr scan, and a matched re-scan with $\text{soft} = \Delta r$, keep large out-of-bracket offsets at several amplitudes but do not yield a clean monotonic amplitude law. The results show that a changing prescribed density hole loads the feed differently from static thickness sampling in this engine. Song/Tu-type works motivate that real RF sheaths move; we do not treat their analytics or PIC reactance as a numerical scoreboard.

Plain Language Summary

Antennas in space plasmas sit inside a low-density sheath. In strong radio-frequency drive that sheath can breathe—growing and shrinking each cycle—instead of staying a fixed gap. We force that breathing as a known function of time inside a three-dimensional fluid electromagnetic code and compare the antenna voltage and current to cases with a fixed gap. The breathing cases look electrically different from the fixed gaps, even after we smooth the numerical sheath edge. That means sheath motion itself can matter for how the antenna loads, before any self-consistent charging model is added.

1 Introduction

A depleted sheath between a metallic antenna and the ambient plasma inserts a largely reactive series impedance and can dominate feed behavior at ionospheric and magnetospheric radio frequencies (Balmain, 1964; Liu et al., 2025; Song et al., 2007; Tu et al., 2008). Probe-oriented models often treat that layer as a quasi-steady coaxial (or related) capacitance in series with the plasma impedance (Liu et al., 2025). High-voltage

transmitting antennas, by contrast, are expected to develop a sheath whose outer radius moves with the drive (Song et al., 2007; Tu et al., 2008).

Three-dimensional plasma fluid FDTD (PF-FDTD) can already impose a *static* volumetric density hole of width S_d around a thin-wire dipole and measure feed impedance (Ward, 2006). Companion campaign work showed strong series-capacitance loading once the hole was seeded from the conductor surface (static sheath capacitance study in preparation). Varying static S_d samples different quasi-steady thicknesses; it does not test whether *time dependence* of the hole changes the feed beyond those thicknesses.

This paper asks a physics question about consequences of motion, not a reproduction of Song or Tu numerics: if $r_s(t)$ is *prescribed* to oscillate at the drive frequency inside the same Maxwell–fluid core, do feed diagnostics differ from static jackets at r_{s0} and $r_{s0} + \Delta r$? A positive answer means dynamic depletion is electromagnetically consequential in this solver class. We do not claim self-consistent charging, orbit-limited collection, or an electron-free ion sheath—only a kinematic density hole.

2 Model and Methods

2.1 Plasma fluid FDTD

We advance Maxwell’s equations on a staggered Yee grid coupled to multi-species fluid continuity and momentum equations following Ward (Ward, 2006). Plasma–Maxwell coupling uses a conductivity mask SIG; conductors use permittivity flags $\text{ER}_* = 0$. Feed voltage and current are recorded at the dipole gap. Unless noted, $f_p = 2 \text{ MHz}$ and $\Delta x = 0.04 \text{ m}$ match the companion static-sheath campaign.

2.2 Kinematic sheath radius

Ambient density $N_0(\mathbf{r})$ is depleted inside a neighborhood of the PEC antenna measured by a PEC-seeded integer distance field. For oscillating runs,

$$r_s(t) = r_{s0} + \Delta r \sin(2\pi f_{\text{osc}} t + \phi), \quad (1)$$

with r_{s0} and Δr in grid cells and f_{osc} defaulting to the RF drive.

Two update policies are compared. *Hard staircase*: $S_d(t) = \text{round}[r_s(t)]$; apply a step density floor when the integer width changes (and floor dynamic density in newly covered cells). *Soft edge (default for later runs)*: blend N_0 over a transition of width `SheathSoftEdge`

Table 1. Hard-staircase pilot phasor Z at 700 kHz (last 50% of each trace).

Case	$\text{Re}\{Z\}$ (Ω)	$\text{Im}\{Z\}$ (Ω)	$ Z $ (Ω)	$\arg Z$ (deg)
Static $r_{s0} = 4$	1.76×10^4	-4.75×10^4	5.06×10^4	-69.7
Static $r_{s0} + 1 = 5$	1.42×10^4	-4.26×10^4	4.49×10^4	-71.6
Oscillating $\Delta r = 1$ (hard)	7.31×10^4	-4.32×10^4	8.49×10^4	-30.6

cells centered on continuous $r_s(t)$, without hard-flooring dynamic N on each update. When $\Delta r = 0$, only the static sheath path runs, preserving the Paper 1-style jacket.

2.3 Experimental design

At each drive frequency we compare static r_{s0} , static $r_{s0} + \lceil \Delta r \rceil$, and oscillating equation (1). Primary observable: fundamental phasor $Z = V/I$ on the last half of each CW record. Success is a clear out-of-bracket difference (dynamic vs static thickness), not a fit to Song/Tu reactance.

Baseline pilots use $f = 700$ kHz, $r_{s0} = 4$, $\Delta r = 1$, $\phi = 0$, and 4.3×10^5 steps. A soft-edge Δr scan extends amplitudes $\{0.5, 1, 1.5, 2\}$ with static brackets through $S_d = 6$.

3 Results

3.1 Hard-staircase pilot

Table 1 and Figure 1 summarize the integer-threshold pilot. Static jackets differ modestly ($|Z_5 - Z_4| \approx 6 \times 10^3 \Omega$). The oscillating case lies far off that segment ($\sim 9 \times$ the bracket span), with a large rise in $\text{Re}\{Z\}$ and a less capacitive $\arg Z$. Late-time current shows strong spikes at radius updates (Figure 2).

3.2 Soft-edge re-test

Repeating the trio with a one-cell soft N_0 blend (Table 2, Figure 3) leaves static controls unchanged and keeps the oscillating phasor well outside the bracket ($\sim 8.4 \times$ the span). $\text{Re}\{Z\}$ and $\arg Z$ shift relative to the hard run, and I_{rms} moves closer to the

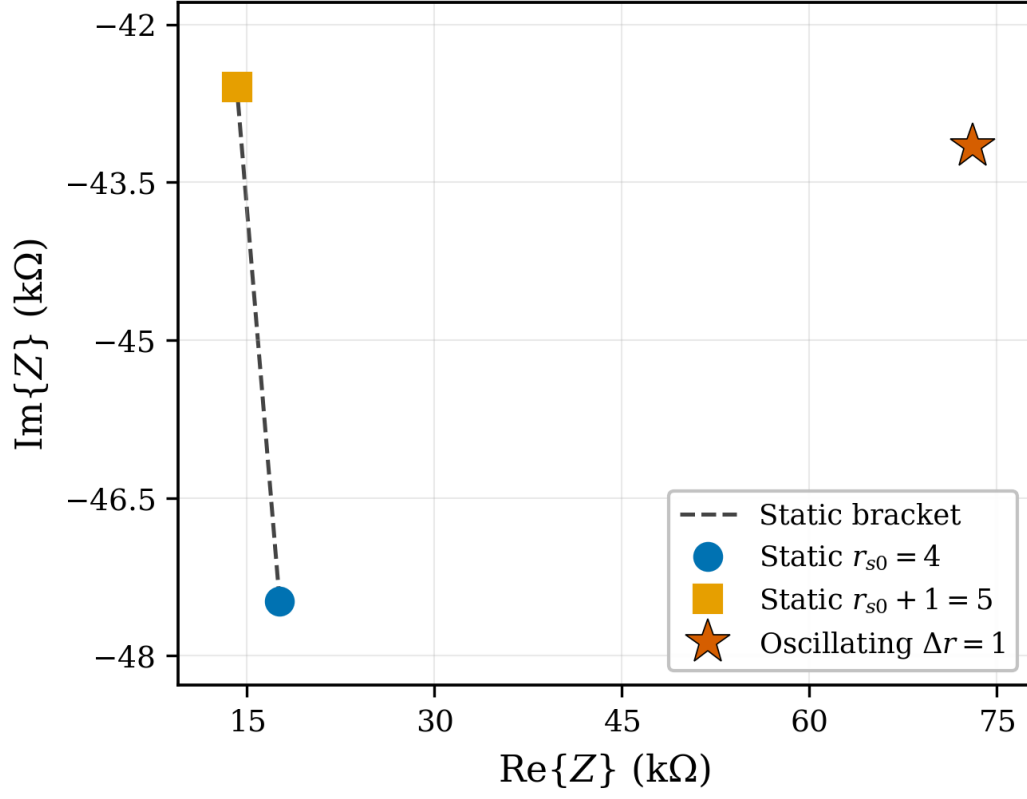


Figure 1. Hard pilot complex-plane phasor Z at 700 kHz: static jackets at $r_{s0} = 4$ and $r_{s0} + 1 = 5$ define the dashed bracket; the oscillating $\Delta r = 1$ case lies far outside that segment.

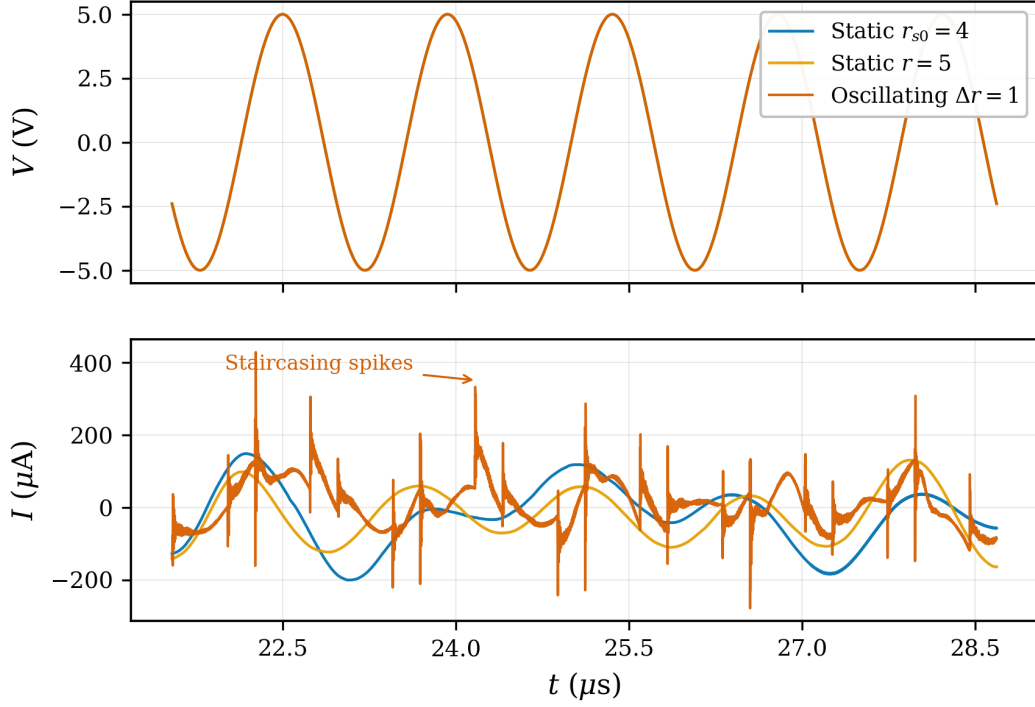


Figure 2. Hard pilot late-time feed traces (last ~ 5 drive cycles). Voltage traces coincide; oscillating current shows impulsive spikes synchronized with integer radius updates.

Table 2. Soft-edge pilot phasor Z at 700 kHz.

Case	$\text{Re}\{Z\}$ (Ω)	$\text{Im}\{Z\}$ (Ω)	$ Z $ (Ω)	$\arg Z$ (deg)
Static $r_{s0} = 4$	1.76×10^4	-4.75×10^4	5.06×10^4	-69.7
Static $r_{s0} + 1 = 5$	1.42×10^4	-4.26×10^4	4.49×10^4	-71.6
Oscillating $\Delta r = 1$ (soft)	6.65×10^4	-5.79×10^4	8.82×10^4	-41.0

static cases, so soft edge reduces impulsive current without removing the qualitative dynamic-loading signature.

3.3 Soft Δr amplitude scan

Soft-edge oscillating runs at $\Delta r \in \{0.5, 1, 1.5, 2\}$ remain far from their static brackets, but the relative offset and $|Z|$ are *not* monotonic in Δr (Figure 4; companion analysis note). Dynamic loading is therefore robust across amplitudes, while a simple $\propto \Delta r$ (or $\propto \dot{r}_s$) scaling is not established.

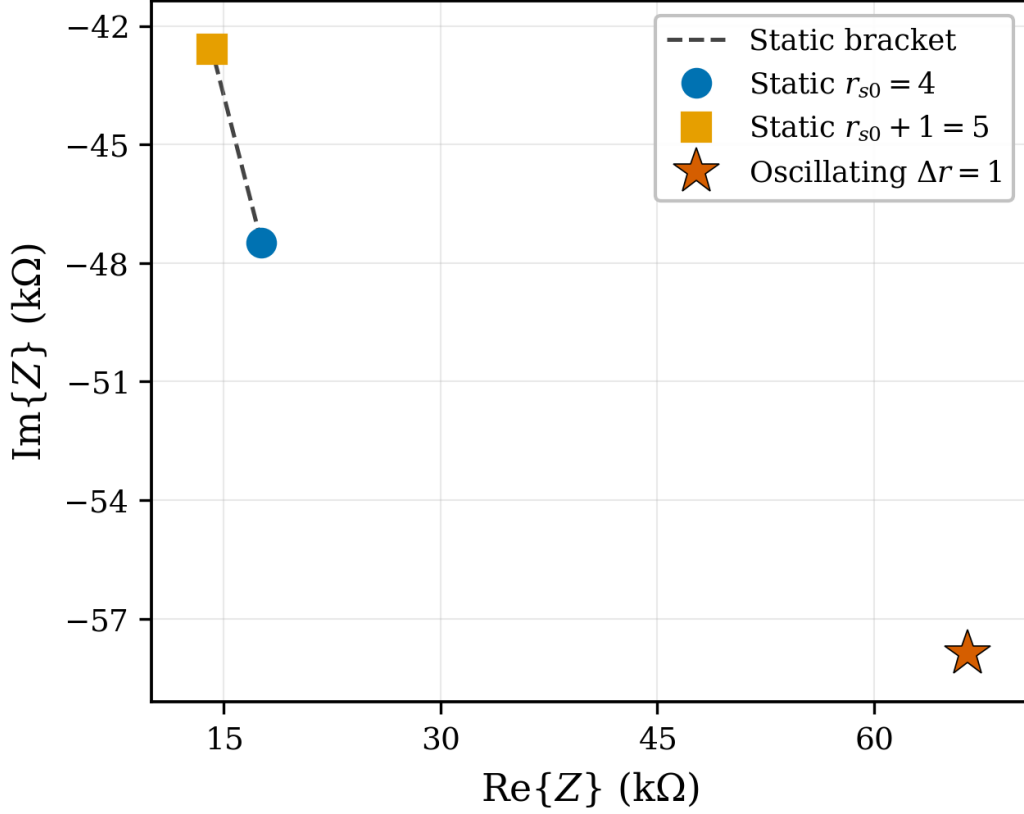


Figure 3. Soft-edge pilot: same static bracket as Figure 1; oscillating $\Delta r = 1$ with one-cell soft blend remains far outside the bracket.

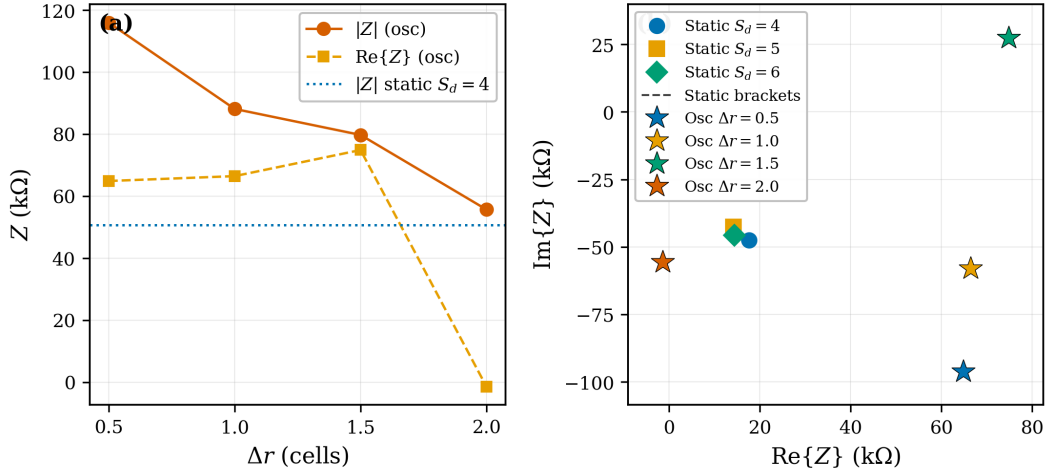


Figure 4. Soft Δr scan at 700 kHz. (a) Oscillating $|Z|$ and $\text{Re}\{Z\}$ versus amplitude, with static $|Z|$ ($S_d = 4$) for reference. (b) Complex-plane loci: static $S_d = 4, 5, 6$ cluster near the origin of the panel; oscillating points remain outside the static brackets and do not follow a monotonic amplitude path.

Table 3. Soft-edge width sensitivity at 700 kHz, $\Delta r = 1$ (static brackets from soft pilot).

Case	edge (cells)	$\text{Re}\{Z\}$	$\text{Im}\{Z\}$	$ Z $	rel. to bracket
Static $r_{s0} = 4$	–	1.76×10^4	-4.75×10^4	5.06×10^4	–
Static $r = 5$	–	1.42×10^4	-4.26×10^4	4.49×10^4	–
Osc	0.5	3.44×10^4	-5.45×10^4	6.44×10^4	3.04
Osc	1.0	6.65×10^4	-5.79×10^4	8.82×10^4	8.37
Osc	2.0	4.34×10^4	-5.89×10^4	7.32×10^4	4.72

Table 4. Sparse soft $\Delta r = 1$ oscillating phasors vs static brackets.

f (kHz)	$\text{Re}\{Z\}$	$\text{Im}\{Z\}$	$ Z $	rel. to bracket
500	1.80×10^4	-2.17×10^4	2.82×10^4	1.22
700	6.65×10^4	-5.79×10^4	8.82×10^4	8.37
1200	-3.50×10^3	-2.75×10^4	2.77×10^4	3.07

3.4 Soft-edge width sensitivity

At fixed $\Delta r = 1$, oscillating runs with $\text{SheathSoftEdge} \in \{0.5, 1, 2\}$ cells all remain far outside the static bracket (relative offsets $\approx 3.0, 8.4, 4.7$; Table 3, Figure 5). Widening the blend moves the complex point and changes I_{rms} (the half-cell edge elevates feed-current RMS), but does not collapse the signature onto the static segment. We therefore retain the one-cell soft edge as the default for sparse- f and phase campaigns.

3.5 Sparse frequency set

Repeating the soft $\Delta r = 1$ protocol at 500 kHz and 1.2 MHz (with the 700 kHz soft pilot) keeps the oscillating phasor outside the static bracket at all three tones (relative offsets $\approx 1.22, 8.37, 3.07$; Table 4, Figure 6). Dynamic loading is therefore not unique to 700 kHz; the strongest offset among these tones remains at 700 kHz.

3.6 Phase scan

At fixed $f = 700$ kHz and $\Delta r = 1$, sweeping the kinematic phase $\phi \in \{0, 45, 90, 135, 180^\circ\}$ rotates $\arg(Z)$ over $\sim 72^\circ$ (Table 5, Figure 7). All five points remain outside the static

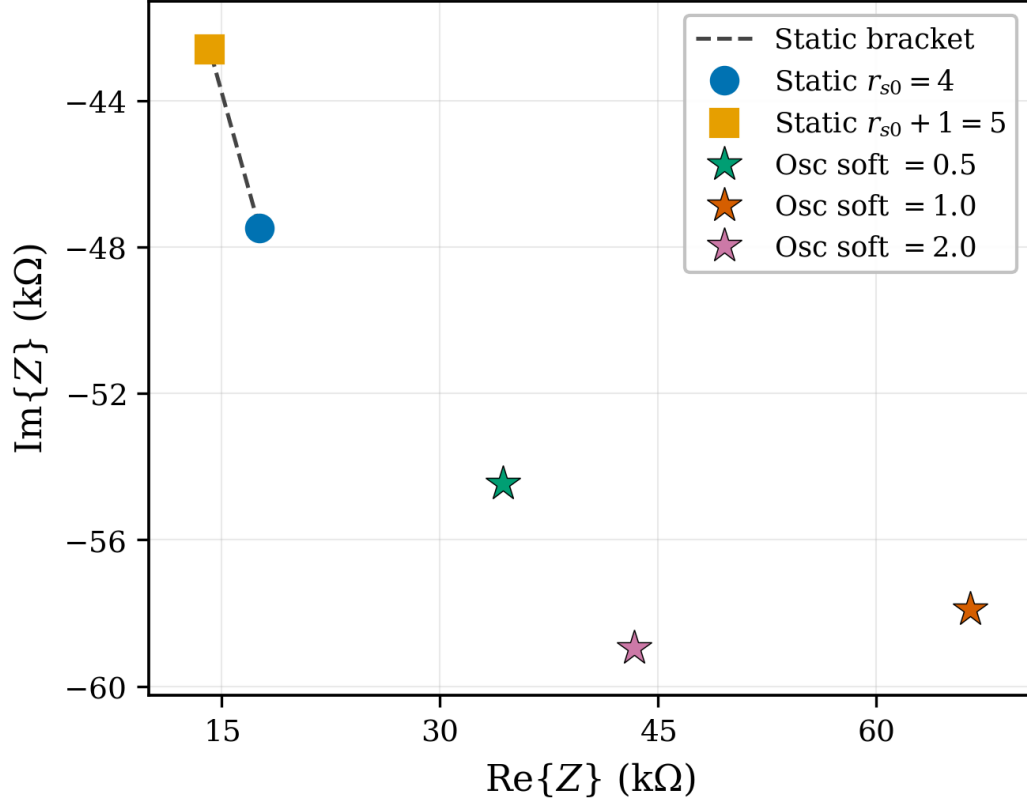


Figure 5. Soft-edge width sensitivity at fixed $\Delta r = 1$, 700 kHz: oscillating Z for blend widths 0.5, 1, and 2 cells remains outside the static bracket.

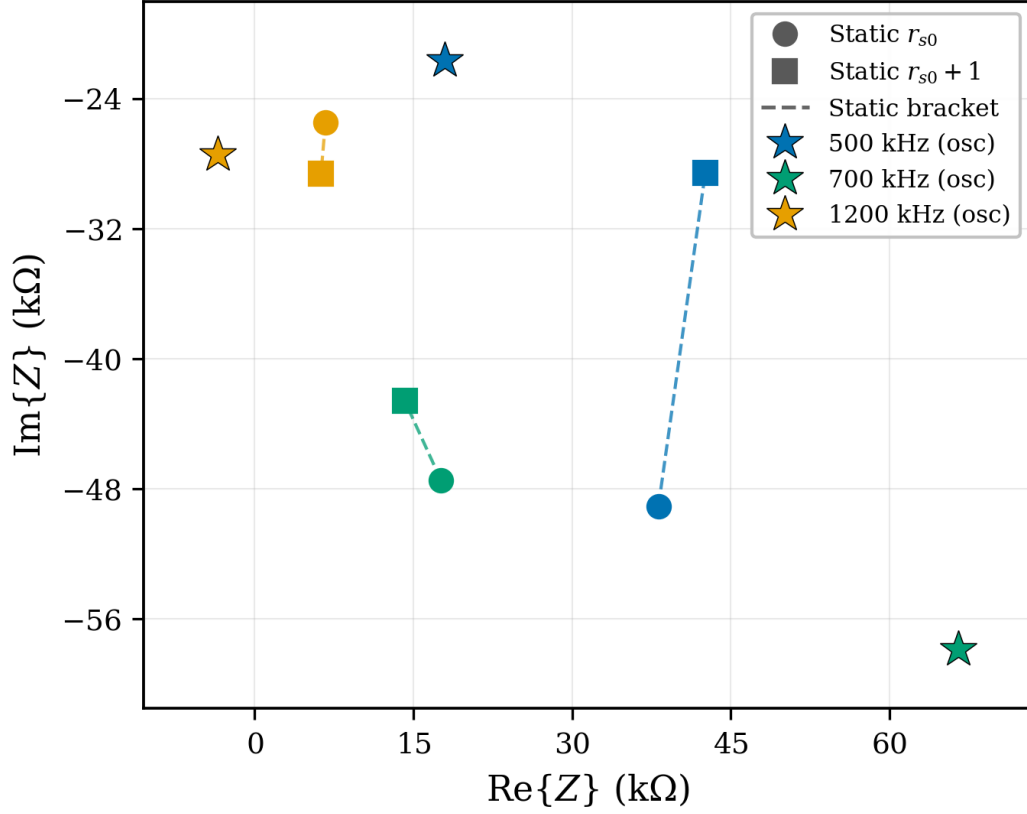


Figure 6. Sparse-frequency soft $\Delta r = 1$ results in the complex plane. Circles/squares and dashed segments are static brackets at each tone; stars are oscillating cases. Color encodes frequency (0.5, 0.7, 1.2 MHz).

Table 5. Soft phase scan at 700 kHz, $\Delta r = 1$.

ϕ (deg)	$\text{Re}\{Z\}$	$\text{Im}\{Z\}$	$\arg Z$ (deg)	rel. to bracket
0	6.65×10^4	-5.79×10^4	-41.0	8.37
45	8.20×10^3	-5.27×10^4	-81.2	1.79
90	3.12×10^4	-1.16×10^4	-20.4	5.92
135	2.39×10^4	-3.07×10^4	-52.0	2.58
180	-1.99×10^3	-4.24×10^4	-92.7	2.71

Table 6. Matched soft Δr scan at 700 kHz (soft = Δr).

Δr	soft	$\text{Re}\{Z\}$	$\text{Im}\{Z\}$	rel. to bracket
0.5	0.5	2.02×10^4	1.07×10^4	8.99
1.0	1.0	6.65×10^4	-5.79×10^4	8.37
1.5	1.5	-7.16×10^3	-3.80×10^4	6.31
2.0	2.0	2.44×10^4	-6.01×10^3	11.24

bracket. The systematic feed-phase response to boundary phase favors coherent moving-boundary coupling over an amplitude-only numerical artifact, while the size of the out-of-bracket offset itself remains ϕ -dependent.

3.7 Matched soft Δr re-scan

Repeating the amplitude scan with $\text{SheathSoftEdge} = \Delta r$ (removing soft-vs-amplitude mismatch) keeps all four oscillating points outside their static brackets (relative offsets $\approx 9.0, 8.4, 6.3, 11.2$; Table 6, Figure 8). Absolute mid-bracket distance and $|Z|$ remain non-monotonic. Matched soft numerics therefore do not restore a simple $\propto \Delta r$ law; non-monotonic amplitude response is reported as a finding.

4 Discussion

Hard and soft pilots reject the null that kinematic $r_s(t)$ is indistinguishable from static sampling of r_{s0} and $r_{s0} + \Delta r$ at this tone. That is the changing-versus-static physics question for a prescribed density hole in 3D PF-FDTD. A sparse frequency set confirms

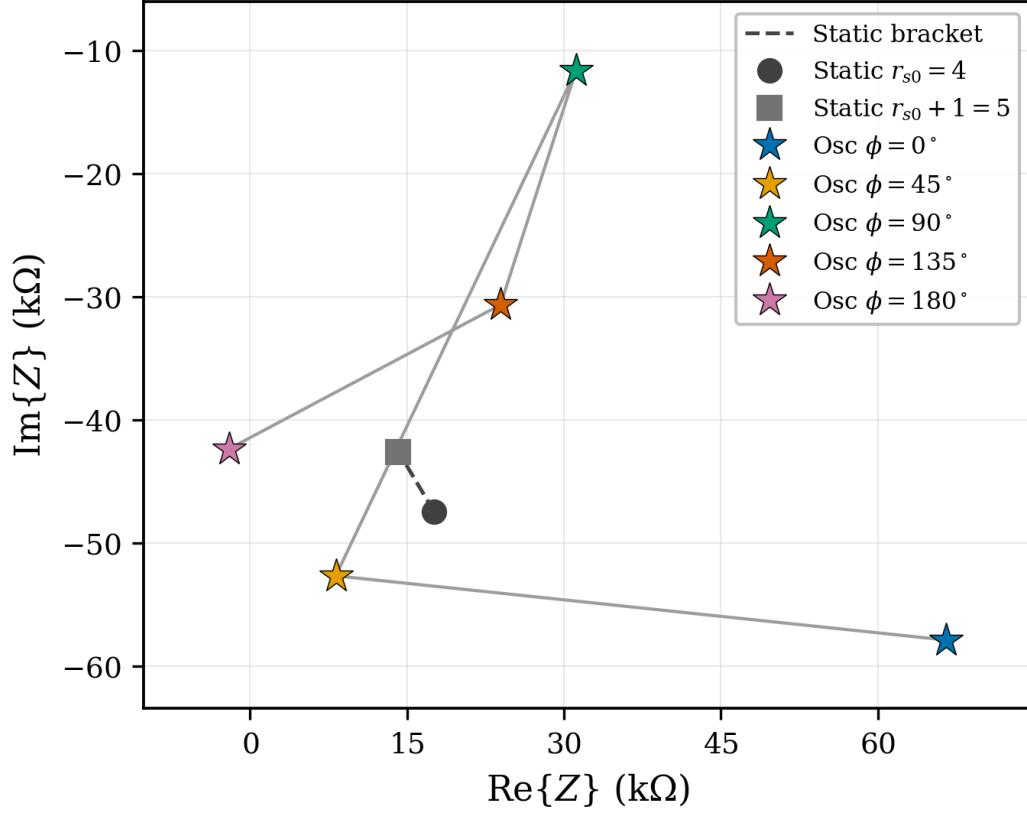


Figure 7. Soft phase scan at 700 kHz, $\Delta r = 1$. Gray markers and dashed segment: static bracket. Colored stars: oscillating cases at $\phi = 0^\circ$ – 180° , connected in phase order. All oscillating points remain outside the bracket while $\arg(Z)$ rotates.

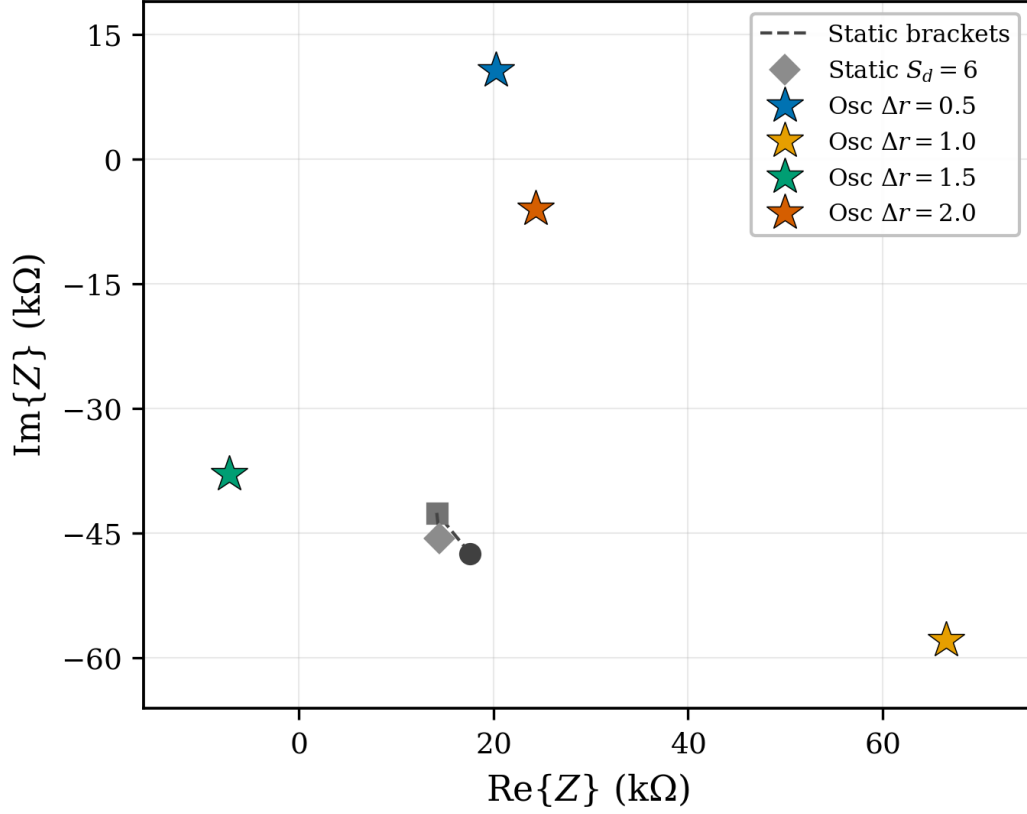


Figure 8. Matched soft Δr scan (soft = Δr) at 700 kHz. Dashed segments: static brackets through $S_d = 6$. Stars: oscillating amplitudes 0.5–2. All remain out-of-bracket without a monotonic amplitude progression.

the same qualitative conclusion at 500 kHz and 1.2 MHz. A phase scan shows $\arg(Z)$ tracks the boundary phase over tens of degrees, consistent with coherent dynamic coupling.

Caveats are first-class, not footnotes. (i) Motion is imposed, not caused by charge—we test consequences of a moving hole, not sheath formation. (ii) Soft $I(t)$ is cleaner than hard staircase but still not purely sinusoidal; part of any single phasor may reflect residual mask updates. (iii) Both fixed-soft and matched-soft Δr scans are non-monotonic; amplitude physics remains open as a documented finding. (iv) Soft-edge width moves the oscillating point without removing the out-of-bracket conclusion at $\Delta r = 1$. (v) The depleted region is a kinematic density floor, not a kinetic electron-free ion sheath (Song et al., 2007; Tu et al., 2008).

Song (Song et al., 2007) and Tu et al. (Tu et al., 2008) motivate that RF sheaths move and that dynamics can matter; they are not used here as a fitting target. Optional next controls are frozen- r_s snapshots and effective-radius static nulls—not a jump to self-consistent charging unless those controls stay coherent.

5 Conclusions

A prescribed oscillating density hole in three-dimensional PF-FDTD produces feed impedance clearly outside static thickness brackets at 700 kHz, for both hard-staircase and soft-edge updates. Soft-edge width sensitivity, a sparse frequency set (0.5–1.2 MHz), a phase scan ($\arg Z$ span $\sim 72^\circ$), and matched soft Δr scans keep that qualitative conclusion and favor coherent boundary-driven loading. Amplitude scaling with sheath swing remains non-monotonic even under matched soft edges. Changing sheath geometry is therefore electromagnetically consequential in this model class; optional stronger nulls can further separate motion from instantaneous thickness sampling.

Acknowledgments

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Open Research

Inputs and scripts live under `projects/tu2008-sheath/` (binary `pffdt_parallel_rs_t`; pilots `run_paper2_rs_t_pilot.ps1`, `run_paper2_delta_r_scan.ps1`; analyzers `analyze_paper2_pilot.py`,

analyze_paper2_delta_r.py). Derived products are under paper2_oscillating_sheath_boundary/analysis/.
Public archival deposit will accompany journal submission.

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